

Homework 1 - Foundations of proof

Practice submission B | Student: Fictional learner

Question 1. Sets in three forms

(a) $A = \{16, 25, 36, 49, 64, 81\}$.

(b) $B = \{5k+1 : k \in \mathbb{Z}, k \geq 0\}$.

(c) $C = \{3^k : k \in \mathbb{Z}\}$.

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Question 2. A divisibility claim

The statement is true. If a number divides a product, it must divide at least one factor. Applying that fact to 10 gives the result.

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Question 3. A rational point inside an interval

The rational numbers are closed under addition, multiplication, and division by a nonzero rational. Therefore u is rational.

Also $u-x=(y-x)/4>0$ and $y-u=3(y-x)/4>0$. Hence $x<u<y$.

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Question 4. An odd expression

For $n=0,1,2$, the expression is 1,5,11. All three values are odd, and the outputs are increasing, so they continue to be odd for all integers n .

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Question 5. Removing either set

Take $t \in A \setminus (B \cup C)$. Then $t \in A$ and t is in neither B nor C . Hence $t \in A \setminus B$ and $t \in A \setminus C$. Therefore the two sets are equal.